



Research paper

Bandgap-oriented topology optimization of cylindrical array breakwaters for broadband wave reflection

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ABSTRACT

Wave attenuation by periodic coastal structures is closely related to bandgap phenomena induced by wave–structure interactions. This study develops a bandgap-oriented topology-optimization framework for cylindrical array breakwaters to achieve broadband and directionally robust wave reflection. A two-dimensional band-structure model based on Bloch theory is first established to characterize wave propagation in infinite periodic arrays and is validated against a frequency-domain potential-flow model for finite arrays. Building on this framework, a bidirectional evolutionary structural optimization (BESO) method is employed to systematically configure the unit-cell topology to maximize the relative bandgap width. The results show that the optimized unit-cell configurations can significantly enlarge bandgaps compared with conventional circular and square cylinders, while also enhancing directional robustness. By combining unit cells optimized for different band indices, multiple bandgaps are effectively merged into a broadband stopband covering a frequency interval from 3.00 to 8.45 rad/s. Hydrodynamic simulations further confirm that the combined configuration achieves near-complete wave reflection with negligible transmission across the entire target frequency range. The proposed framework establishes a direct link between bandgap engineering and wave-attenuation performance, providing a systematic and physically grounded design strategy for compact and efficient coastal breakwaters.

1. Introduction

Coastal protection remains a fundamental challenge in ocean and coastal engineering, particularly in regions exposed to energetic and highly variable wave climates. Conventional breakwaters (Faraci et al., 2015), such as rubble-mound and gravity-based structures, are widely used because of their proven effectiveness in reducing wave energy. However, these structures typically require large volumes of material, substantial seabed occupation, and high construction and maintenance costs, which limit their applicability in shallow or environmentally sensitive coastal zones. As a lightweight and modular alternative, arrays of bottom-mounted vertical cylinders have received increasing interest for wave attenuation (Ji and Suh, 2010; Anh et al., 2025; Peng et al., 2025), owing to their structural simplicity, adaptability, and relatively small environmental footprint. Related interest in the hydrodynamic performance of marine structure arrays, including those integrating energy-conversion functionality, has also been reported (Cheng et al., 2022, 2024).

To analyze wave interactions with bottom-mounted cylinder arrays, most studies in coastal engineering have relied on linear potential-flow theory formulated in the frequency domain. Under the assumptions of inviscid, incompressible flow and small-amplitude waves, the wave–structure interaction problem reduces to a Helmholtz equation (Linton and Evans, 1990; Maniar and Newman, 1997; McIver, 2000; Zhou and Song, 2025) for the complex velocity potential $\phi(x, y)$, defined on the horizontal plane O - xy , as shown in Fig. 1:

$$\nabla^2 \phi(x, y) + k^2 \phi(x, y) = 0, \quad (1)$$

where k is the wave number, satisfying the linear dispersion relation with incident wave frequency ω , given by $\omega^2 = gk \tanh(kh)$, where g is the gravitational acceleration, and h denotes the still-water depth. The free-surface elevation η is subsequently obtained from the velocity potential as $\eta = i\omega \phi(x, y)/g$, with $i^2 = -1$. Within this framework, the spatial distribution of the free-surface elevation, together with the reflection and transmission characteristics of finite cylinder arrays, can

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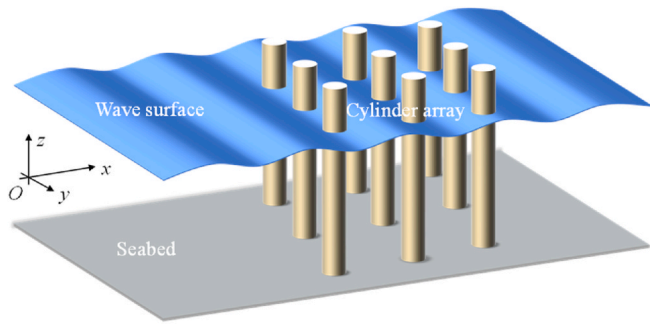


Fig. 1. Schematic of wave interactions with bottom-mounted cylindrical array breakwaters in open water.

be evaluated efficiently. As a result, this approach has been widely used to assess the hydrodynamic performance of periodic and quasi-periodic breakwater systems (Zhou and Song, 2025).

However, linear potential-flow theory in the frequency domain has inherent limitations when applied to periodic arrays of bottom-mounted cylinders. In particular, wave attenuation performance is typically evaluated through frequency-by-frequency scattering calculations for finite arrays, making it difficult to identify global wave-blocking characteristics over broad frequency ranges. Moreover, the predicted reflection and transmission properties depend on the finite length of the array and the associated boundary effects, which may obscure the intrinsic wave-propagation mechanisms governed by the underlying periodicity. Consequently, although frequency-domain potential-flow models are effective for quantifying hydrodynamic responses, they offer limited physical insight into why certain frequency ranges exhibit strong wave attenuation and whether such behavior is robust with respect to wave direction and array configuration.

From a physical perspective, the wave attenuation performance of periodic cylinder arrays is closely linked to their dispersive properties (McIver, 2000). When surface waves interact with a periodic arrangement of obstacles, strong multiple scattering may occur, leading to frequency ranges in which wave propagation is inhibited (Porter and Porter, 2003; Liu et al., 2019a; Song et al., 2026b; among others). These frequency intervals, commonly referred to as bandgaps, are associated with Bragg scattering mechanisms and correspond to regimes in which no propagating Bloch modes exist within the periodic medium. Within such bandgaps, incident wave energy is predominantly reflected rather than transmitted, giving rise to pronounced wave-blocking behavior that is largely independent of array length. In this context, band-structure analysis provides a powerful theoretical framework for understanding and predicting wave-blocking behavior in periodic marine structures.

Similar concepts have been extensively explored in the fields of phononic and photonic crystals, where periodic structures are deliberately engineered to tailor wave dispersion and open wide bandgaps for applications such as vibration isolation and wave filtering. Inspired by these advances, bandgap-based design concepts have gradually been introduced into coastal and ocean engineering. Representative studies include investigations of periodic submerged structures or seabed topographies (Mei et al., 1988; Hu et al., 2003a; Liu et al., 2019b; Ning et al., 2022), periodic arrays of vertical cylinders (Linton and Evans, 1990; Maniár and Newman, 1997; McIver, 2000; Hu et al., 2003b; Jeong et al., 2004), and floating body arrays (Carter and McIver, 2012; Song et al., 2026b). Bandgap theory establishes a clear physical connection between the dispersion characteristics of idealized infinite periodic arrays and the wave attenuation performance of finite breakwater systems. Based on this framework, several representative wave phenomena have been identified and interpreted, including rainbow trapping and reflection (Wilks et al., 2022; Xu et al., 2023; Xie et al., 2025; Zhou and Song, 2025). Nonlinear time-domain analyses (Song et al., 2025c; Song

et al., 2025d; Song and Jin, 2026; Song et al., 2026c; Zhou and Song, 2026) can provide deeper insights that complement linear band-structure analysis. Thus, band-structure analysis offers a systematic and physically transparent approach for characterizing the intrinsic dispersive properties of periodic cylinder arrays and for predicting frequency intervals associated with strong wave attenuation.

Despite these advances, most existing studies on bandgap phenomena in periodic cylinder arrays have focused on analyzing wave-propagation characteristics for prescribed or idealized geometries. In many cases, the cylinder shape, size, and spatial arrangement are fixed a priori, and band-structure analysis is used primarily as a diagnostic tool to identify bandgaps associated with a given configuration. Although such studies have significantly improved the physical understanding of wave-array interactions, they offer limited guidance on how to systematically design or optimize the structural geometry to enhance wave attenuation performance from an engineering perspective. In particular, the deliberate optimization of bandgaps, especially their width, location, and robustness, remains largely unexplored in the context of coastal breakwater design. Although fluid resonance (Song et al., 2021, 2025a, 2025b; Zhao et al., 2022) has recently been introduced into breakwater systems to enhance performance against low-frequency, long-period waves, such mechanisms are generally less effective against practical multi-frequency irregular waves. In practical applications, effective coastal protection often requires strong wave attenuation over a broad frequency range rather than at isolated frequencies. However, bandgaps arising from simple periodic arrangements are typically narrow and sensitive to geometric parameters, wave direction, and water depth, making broadband wave blocking a persistent challenge. As a result, translating bandgap concepts into engineering-relevant designs that achieve broadband reflection and maintain robustness under realistic wave conditions remains an open problem.

These limitations highlight the need for a systematic design framework that directly links band-structure characteristics with structural optimization. Rather than passively analyzing the dispersive properties of prescribed geometries, such a framework should actively optimize the geometry and topology of periodic cylinder arrays to maximize bandgap width and achieve broadband wave attenuation, thereby advancing bandgap-based concepts toward practical coastal protection applications. Motivated by these considerations, and inspired by bandgap-maximization principles widely explored in phononic and photonic crystals, this study develops a systematic optimization framework for periodic bottom-mounted cylinder arrays. To the best of the authors' knowledge, however, such concepts have rarely been translated into the design of coastal breakwater systems composed of bottom-mounted cylinder arrays. Unlike previous studies that focused on optimizing predefined geometric parameters (Somervell et al., 2017), the present study formulates the problem from an inverse-design perspective, in which the structural topology is actively tailored to achieve the desired broadband wave-attenuation performance. By integrating band-structure analysis with topology optimization, the proposed approach provides a direct and physically grounded design methodology for engineering efficient periodic breakwaters. Future designs may also benefit from considering hybrid systems that combine wave attenuation with energy conversion, as explored in recent studies of flexible and multimode wave energy converters (Song et al., 2025d; Cheng et al., 2026).

Following this introduction, Section 2 presents the bandgap theory for wave interactions with infinite arrays of bottom-mounted cylinders, which establishes the theoretical foundation for the topology optimization conducted in this study. Section 3 introduces the BESO-based topology optimization framework, in which bandgap width is formulated as the objective function, and the associated constraints and numerical implementation are described in detail. Section 4 presents a parameter-sensitivity analysis to investigate the influences of key geometric and physical parameters on bandgap formation, thereby

providing guidance for the optimization strategy. Section 5 presents the main results, focusing on bandgap maximization through topology optimization and its translation into enhanced wave attenuation performance in finite cylinder arrays, including broadband and directionally robust wave-blocking behavior. Finally, Section 6 summarizes the main findings and discusses their implications for the engineering design of efficient periodic breakwater systems.

The main novelties of this study can be summarized as follows. First, unlike conventional approaches that rely on predefined geometries or parametric optimization, the present study formulates the design problem from an inverse-design perspective, in which the structural topology is actively tailored to achieve the desired bandgap characteristics. Second, a bandgap-oriented topology-optimization framework is developed to enable the systematic maximization of bandgap width at the unit-cell level. Third, a direct and quantitative link is established between band-structure characteristics and the wave attenuation performance of finite cylinder arrays. Finally, a bandgap-combination strategy is proposed to overcome the inherent narrowband limitation of single-topology designs, thereby achieving broadband and directionally robust wave blocking. The proposed framework aims to bridge the gap between dispersion-based theoretical analysis and engineering-oriented breakwater design.

2. Bandgap theory for wave interactions with infinite bottom-mounted cylindrical arrays

The present study is based on a linear potential-flow framework for water waves, in which the fluid is assumed to be inviscid, incompressible, and irrotational, and wave propagation is described under small-amplitude wave theory. The structures are assumed to be rigid and fixed, and periodic boundary conditions are applied to represent an infinite cylindrical array. These assumptions are adopted to enable tractable band-structure analysis and topology optimization while capturing the essential physics of wave-structure interactions. Within this framework, the cylindrical array breakwater consists of periodically arranged structural units with a characteristic length scale. The optimized unit-cell topology governs the scattering and interference of incident waves, leading to the formation of bandgaps that inhibit wave propagation. From an engineering perspective, such configurations can be interpreted as compact modular breakwater systems, in which the arrangement and geometry of individual units can be systematically tailored to achieve broadband wave attenuation.

Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be two independent two-dimensional vectors spanning the lattice, which define the translational symmetry of the mean center positions of the cylinders in the horizontal plane. Any lattice translation can be expressed as $\mathbf{R} = m_1\mathbf{a}_1 + m_2\mathbf{a}_2$, where m_1 and m_2 are integers. The corresponding reciprocal lattice vectors are given by $\mathbf{Q} = n_1\mathbf{b}_1 + n_2\mathbf{b}_2$, which satisfy the relation $\mathbf{Q} \cdot \mathbf{R} = 2\pi\Xi$, where Ξ is an integer. Consequently, the direct and reciprocal vectors ($\mathbf{a}_1, \mathbf{a}_2$) and ($\mathbf{b}_1, \mathbf{b}_2$) obey the orthogonality condition $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$, where $i, j = 1, 2$, and δ_{ij} denotes the Kronecker delta. The two-dimensional space spanned by $\mathbf{R}$ corresponds to the real physical space, whereas the space formed by $\mathbf{Q}$ is referred to as the reciprocal space. By invoking Bloch's theorem (Ashcroft and Mermin, 1976), the velocity potential $\phi(x, y)$ in the Helmholtz equation (see Eq. (1)) can be expressed in Bloch form as

$$\phi(\mathbf{r}) = e^{i\mathbf{q} \cdot \mathbf{r}} \psi(\mathbf{r}), \quad (2)$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ is the position vector in real physical space, $\mathbf{q} = q_x\mathbf{i} + q_y\mathbf{j}$ is the Bloch vector in reciprocal space, and $\mathbf{i}$ and $\mathbf{j}$ are unit vectors along the x - and y -directions, respectively. Substituting Eq. (2) into the Helmholtz equation (Eq. (1)) yields the governing equation for the periodic function $\psi(\mathbf{r})$:

$$\nabla^2 \psi(\mathbf{r}) + i\mathbf{q} \cdot \nabla \psi(\mathbf{r}) + \nabla \cdot [i\mathbf{q} \psi(\mathbf{r})] - \|\mathbf{q}\|^2 \psi(\mathbf{r}) = -k^2 \psi(\mathbf{r}). \quad (3)$$

Owing to the spatial periodicity of $\psi(\mathbf{r})$, periodic boundary

conditions are imposed on the boundaries of the lattice domain, such that $\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{R})$. Accordingly, the impermeable boundary condition on the cylinder surface is reformulated as $\partial\psi(\mathbf{r})/\partial\mathbf{n} = 0$. Here, $\mathbf{n}$ denotes the outward unit normal vector on the fluid boundary. For a prescribed Bloch vector $\mathbf{q}$, Eq. (3), together with the boundary conditions $\psi(\mathbf{r}) = \psi(\mathbf{r} + \mathbf{R})$ and $\partial\psi(\mathbf{r})/\partial\mathbf{n} = 0$, is solved numerically using the finite-element method. The lattice domain Ω is discretized into finite computational elements, within which the periodic function is approximated as $\psi(\mathbf{r}) = \mathbf{N}\Psi(\mathbf{r})$, where $\mathbf{N}$ is the shape-function matrix. Applying the standard Galerkin weighted-residual formulation leads to the following weak form:

$$\int_{\Omega} \mathbf{N}^T (\nabla^2 \mathbf{N}\Psi + i\mathbf{q} \cdot (\nabla \mathbf{N}\Psi) + \nabla^T \cdot (i\mathbf{q} \mathbf{N}\Psi) - \|\mathbf{q}\|^2 \mathbf{N}\Psi + k^2 \mathbf{N}\Psi) d\Omega = 0. \quad (4)$$

By applying Green's theorem, the first term on the left-hand side of Eq. (4) can be expressed as

$$\int_{\Omega} \mathbf{N}^T \nabla^2 \mathbf{N}\Psi d\Omega = \oint_{\Gamma} \mathbf{N}^T \nabla(\mathbf{N}\Psi) \cdot \mathbf{n} ds - \int_{\Omega} \nabla \mathbf{N}^T \nabla(\mathbf{N}\Psi) d\Omega. \quad (5)$$

Substituting Eq. (5) into Eq. (4) yields the following algebraic equation:

$$(\mathbf{K} - k^2 \mathbf{M})\Psi = 0, \quad (6)$$

with $\mathbf{K}$, $\mathbf{M}$, $\mathbf{D}$, and $\mathbf{A}$ defined as $\mathbf{K} = \mathbf{D} + i\mathbf{q}(\mathbf{A}^T - \mathbf{A}) + \|\mathbf{q}\|^2 \mathbf{M}$, $\mathbf{M} = \int_{\Omega} \mathbf{N}^T \mathbf{N} d\Omega$, $\mathbf{D} = \int_{\Omega} (\nabla \mathbf{N})^T \nabla \mathbf{N} d\Omega$, and $\mathbf{A} = \int_{\Omega} \mathbf{N}^T \nabla \mathbf{N} d\Omega$.

Equation (6) mathematically constitutes a generalized eigenvalue problem. In principle, solving this problem requires sampling all Bloch wave vectors $\mathbf{q}$ over the entire reciprocal space. However, owing to lattice periodicity, $\mathbf{q}$ can be restricted to the first Brillouin zone, defined as $q_x \in [-\pi/L, \pi/L]$ and $q_y \in [-\pi/W, \pi/W]$ (Ashcroft and Mermin, 1976). Moreover, because the first Brillouin zone exhibits central symmetry, the computational domain can be further reduced to the irreducible Brillouin zone. Compared with interior points, boundary points of the Brillouin zone possess higher symmetry, and extrema of the wave-number $k(\mathbf{q})$ occur at high-symmetry points, including $\Gamma = (0, 0)$, $X = (\pi/L, 0)$, and $M = (\pi/L, \pi/W)$. As a result, the solutions at interior points of the Brillouin zone are bounded by the frequency range determined by these high-symmetry points. The Bloch vector $\mathbf{q}$ can therefore be sampled along the boundary of the irreducible Brillouin zone, following the standard path $\Gamma-X-M-\Gamma$. According to bandgap theory (Ashcroft and Mermin, 1976), the segments of the $\Gamma-X-M-\Gamma$ path correspond to specific wave-propagation directions in physical space. Specifically, the $\Gamma-X$ direction represents wave propagation along the x -direction ($\mathbf{r} = L\mathbf{i} + 0\mathbf{j}$), the $M-\Gamma$ direction corresponds to diagonal propagation ($\mathbf{r} = L\mathbf{i} + W\mathbf{j}$), and the $X-M$ direction represents an intermediate propagation direction between the two.

For a given Bloch wave vector $\mathbf{q}$, solving Eq. (6) yields a series of eigenvalues and corresponding eigenvectors. The eigenvalues are ordered in ascending sequence to obtain the wavenumbers $k_n(\mathbf{q})$ and the associated periodic functions $\Psi_n(\mathbf{r})$. In accordance with Bloch's theorem (Eq. (2)), the periodic functions $\Psi_n(\mathbf{r})$ are used to construct the corresponding wave-propagation modes, while the dispersion relations $k_n(\mathbf{q})$ collectively form the band structure. For convenience, the wavenumbers $k_n(\mathbf{q})$ are converted into angular frequencies $\omega_n(\mathbf{q})$ via the linear wave-dispersion relation.

To validate the capability of the band-structure model in predicting full-reflection frequency intervals, that is, bandgaps, a diffraction problem involving an array of eight equally spaced bottom-mounted cylinders is considered. Benchmark solutions are provided by a fully validated two-dimensional linear potential-flow model in the frequency domain (Zhou and Song, 2025). As shown in Fig. 2(a), the two-dimensional numerical wave tank has a length of $20L$ and a width of $W = 2.20$ m. Here, L denotes the spacing between adjacent cylinders. Two representative spacings, $L = 1.20$ m and 2.20 m, are examined. The

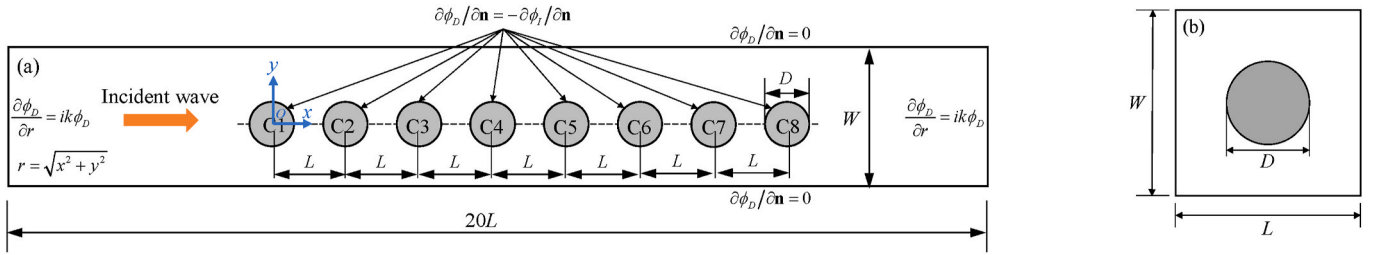


Fig. 2. (a) Schematic plan view of eight bottom-mounted cylinders in the numerical wave tank, and (b) schematic of the lattice in the bandgap-theory framework.

cylinder diameter is $D = 1.00$ m, and the still-water depth is $h = 0.70$ m. The incident wave propagates along the horizontal x -axis, corresponding to the Γ - X direction of the irreducible Brillouin zone. A schematic of the corresponding lattice, with length L and width W , is provided in Fig. 2(b). Details of the numerical implementation of the two-dimensional linear potential-flow model are provided in Appendix A.

Comparisons between the frequency intervals of the bandgaps in the Γ - X direction, calculated by the band-structure model, and the full-reflection condition ($|R|^2 = 1$ and $|T|^2 = 0$), simulated by the frequency-domain model, are shown in Fig. 3. The results demonstrate that the established band-structure model is capable of accurately predicting full-reflection frequency intervals through the identification of bandgaps. Specifically, for the case of $L = 1.20$ m (Fig. 3(a)), two bandgaps, that is, $\omega \in [4.29, 5.14 \text{ rad/s}] \cup [5.58, 6.06 \text{ rad/s}]$, show excellent agreement with the full-reflection frequency intervals obtained from the frequency-domain model. For the case of $L = 2.20$ m (Fig. 3(b)), four bandgaps, that is, $\omega \in [2.70, 3.56 \text{ rad/s}] \cup [5.46, 5.68 \text{ rad/s}] \cup [5.87, 6.27 \text{ rad/s}] \cup [6.50, 6.70 \text{ rad/s}]$, are similarly validated. More importantly, the band-structure model offers a substantial computational advantage. For the representative case of $L = 1.20$ m, the linear frequency-domain potential-flow model requires solving a matrix of size $80,733 \times 80,733$, with a computational cost of approximately 10.5 h. In contrast, the band-structure model reduces the matrix size to 1740×1740 and the computational time to about 8 min. This efficiency provides a solid foundation for the subsequent topology optimization. Although the band-structure model enables rapid identification of full-reflection frequency ranges, the bandgaps obtained for the simple cylindrical configurations considered here are not necessarily optimal and can be further enhanced through topology optimization.

3. Bidirectional evolutionary structural optimization

Motivated by recent advances in phononic and photonic crystals, in which bandgaps are deliberately engineered through topology optimization to achieve vibration attenuation and wave filtering, this study extends the concept of bandgap maximization to coastal engineering

applications. Unlike conventional parametric designs based on simple geometric shapes, the present study develops an optimization framework for arrays of bottom-mounted vertical cylinders, in which the cross-sectional topology of individual cylinders is systematically optimized at the unit-cell level to maximize wave bandgaps over prescribed frequency ranges. Furthermore, optimized unit-cell topologies targeting different bandgaps are combined at the array level to further broaden the effective bandgap, thereby enabling broadband and direction-independent full wave reflection, which is particularly desirable for practical breakwater applications.

3.1. Design variables, objective function, and constraints

The bidirectional evolutionary structural optimization (BESO) method (Yang et al., 1999) is employed in this study because of its ability to simultaneously remove and add material during each iteration, thereby enhancing numerical robustness and convergence efficiency compared with conventional element-removal-based approaches. This feature is particularly advantageous for bandgap optimization problems, in which both material distribution and connectivity play critical roles in wave-scattering behavior. As shown in Fig. 4, the unit cell is discretized into N square finite elements. Each element is assigned a prescribed artificial density ρ_i ($i = 1, 2, \dots, N$, where i is the element index) to characterize the material distribution within the unit cell. Specifically, $\rho_i = 1$ corresponds to a fluid-phase element, $\rho_i = 0$ corresponds to a

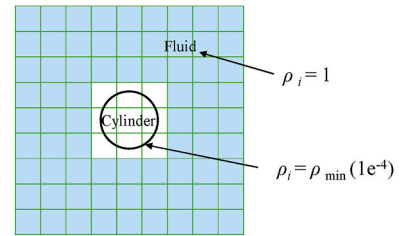


Fig. 4. Schematic of element-density discretization in a lattice.

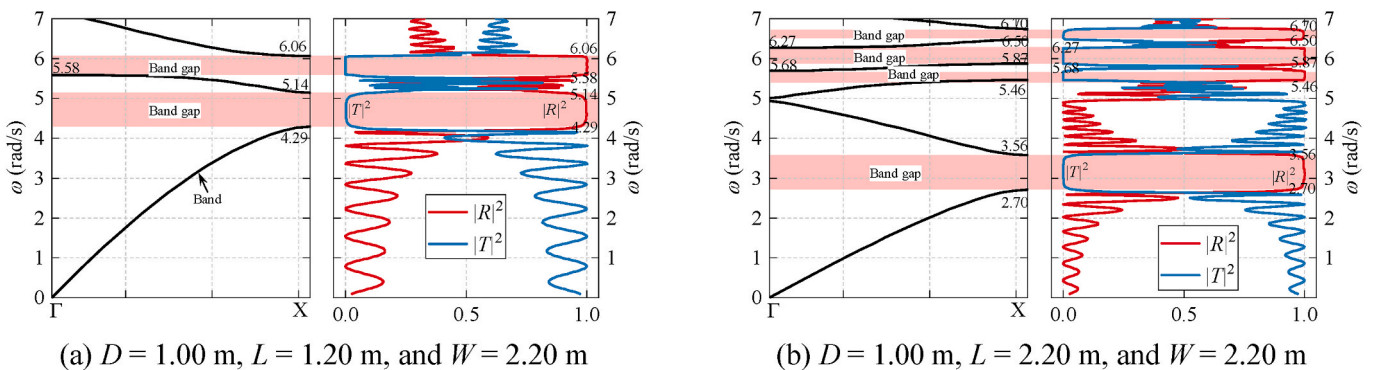


Fig. 3. Comparisons of the band structure along Γ - X calculated by bandgap theory and the squared reflection coefficient $|R|^2$ and squared transmission coefficient $|T|^2$ at different incident wave frequencies ω obtained by the linear frequency-domain model.

solid-phase (cylinder) element, and intermediate values, that is, $0 < \rho_i < 1$, correspond to transition elements during the optimization process. To avoid numerical singularities, a small positive value $\rho_{\min} = 1 \times 10^{-4}$, instead of 0, is assigned as the minimum element density. Consequently, the spatial distribution of the density function ρ_i fully defines the geometric topology of the cylinder cross-section and serves as the primary design variable. During the optimization process, the bandgap of the unit-cell structure is maximized by iteratively adjusting the spatial distribution of the density function ρ_i .

For a given unit-cell configuration, bandgaps naturally arise in the corresponding band structure. The objective of the present optimization is to maximize the width of a targeted bandgap. Following the band-structure formulation introduced in Section 2, as shown in Fig. 3, the bandgap between the n th band $\omega_n(\mathbf{q})$ and the $(n+1)$ th band $\omega_{n+1}(\mathbf{q})$ is referred to as the n – $(n+1)$ bandgap; for example, the 1–2 bandgap refers to the gap between $\omega_1(\mathbf{q})$ and $\omega_2(\mathbf{q})$. The absolute width of this bandgap, $\Delta\omega$, is defined as

$$\Delta\omega = \min[\omega_{n+1}(\mathbf{q})] - \max[\omega_n(\mathbf{q})], \quad (7)$$

where $\min[\]$ and $\max[\]$ denote the minimum and maximum values of the respective bands. Because the absolute bandgap width $\Delta\omega$ depends on the characteristic length scale of the unit cell, the relative bandgap width, $\Delta\omega/\omega_{\text{mid}}$, provides a more meaningful and scale-independent performance measure (Meng et al., 2015). Accordingly, the objective function $f(\rho_i)$, expressed in terms of the relative bandgap width, is defined as

$$f(\rho_i) = \left\{ \frac{\Delta\omega}{\omega_{\text{mid}}} \right\} (\rho_i) = \frac{\min[\omega_{n+1}(\mathbf{q})] - \max[\omega_n(\mathbf{q})]}{\left\{ \frac{\min[\omega_{n+1}(\mathbf{q})] + \max[\omega_n(\mathbf{q})]}{2} \right\}} = 2 \frac{\min[\omega_{n+1}(\mathbf{q})] - \max[\omega_n(\mathbf{q})]}{\min[\omega_{n+1}(\mathbf{q})] + \max[\omega_n(\mathbf{q})]}, \quad (8)$$

where ω_{mid} denotes the mid-frequency of the target bandgap. The optimization problem is formulated to maximize $f(\rho_i)$.

In addition to the design variables and objective function, two constraints are imposed. First, the element density must satisfy $\rho_{\min} \leq \rho_i \leq 1$. Second, a volume-fraction constraint is introduced to prevent trivial solutions in which the unit cell is either fully solid or fully fluid. Specifically, the fluid-phase volume fraction is constrained as

$$V(\rho_i) = \sum_{i=1}^N \rho_i V_i \Big/ V_{\text{total}} = V^*, \quad (9)$$

where V_{total} denotes the total volume of the unit cell, V_i is the volume of element i , Σ denotes summation over all elements, and V^* is the target control volume fraction. The complete topology-optimization problem is formulated as

$$\begin{cases} \text{Maximize : } f(\rho_i) = \left\{ \frac{\Delta\omega}{\omega_{\text{mid}}} \right\} (\rho_i) = 2 \frac{\min[\omega_{n+1}(\mathbf{q})] - \max[\omega_n(\mathbf{q})]}{\min[\omega_{n+1}(\mathbf{q})] + \max[\omega_n(\mathbf{q})]} \\ \text{Subject to : } V^* = V(\rho_i) = \sum_{i=1}^N \rho_i V_i \Big/ V_{\text{total}} \\ \rho_i \in [\rho_{\min}, 1] \end{cases} \quad (10)$$

3.2. Iterative optimization process

In the BESO framework, the unit-cell topology is iteratively updated based on the sensitivity of the objective function with respect to the element-density variables. The sensitivity of the objective function $f(\rho_i)$ with respect to the element density ρ_i , denoted by α_i , is expressed as

$$\alpha_i = \frac{\partial f(\rho_i)}{\partial \rho_i} = 4 \frac{\left[\frac{\partial \min \omega_{n+1}(\mathbf{q})}{\partial \rho_i} \max \omega_n(\mathbf{q}) - \min \omega_{n+1}(\mathbf{q}) \frac{\partial \max \omega_n(\mathbf{q})}{\partial \rho_i} \right]}{[\min \omega_{n+1}(\mathbf{q}) + \max \omega_n(\mathbf{q})]^2}. \quad (11)$$

Here, the derivative of the frequency $\omega(\mathbf{q})$ with respect to ρ_i can be further expressed using the chain rule as

$$\frac{\partial \omega(\mathbf{q})}{\partial \rho_i} = \frac{\partial \omega(\mathbf{q})}{\partial k(\mathbf{q})} \frac{\partial k(\mathbf{q})}{\partial \rho_i}. \quad (12)$$

Here, the derivative of $\omega(\mathbf{q})$ with respect to $k(\mathbf{q})$ is evaluated using the linear wave-dispersion relation, that is, $\omega^2 = gk \tanh(kh)$, as follows:

$$\frac{\partial \omega(\mathbf{q})}{\partial k(\mathbf{q})} = \frac{g \tanh kh + gk(1 - \tanh^2 kh)h}{2\sqrt{gk \tanh kh}}. \quad (13)$$

The second term, $\partial k(\mathbf{q})/\partial \rho_i$, can be obtained from the eigenvalue equation (Eq. (6)) by explicitly incorporating the density variable into the global stiffness and mass matrices. Specifically, the global stiffness matrix $\mathbf{K}$ and the global mass matrix $\mathbf{M}$ are interpolated as

$$[\mathbf{K}(\rho_i) - k^2 \mathbf{M}(\rho_i)] \boldsymbol{\psi} = 0, \quad (14)$$

with $\mathbf{K}(\rho_i) = \sum_{i=1}^N p_K(\rho_i) \mathbf{k}_i$ and $\mathbf{M}(\rho_i) = \sum_{i=1}^N p_M(\rho_i) \mathbf{m}_i$. Here, $\mathbf{k}_i$ and $\mathbf{m}_i$ are the stiffness and mass matrices of element i , respectively. The penalization functions $p_K(\rho_i)$ and $p_M(\rho_i)$ are introduced to discourage intermediate densities. To mitigate the emergence of localized modes and numerical instabilities, the rational approximation of material

properties (RAMP) interpolation scheme (Stolpe and Svanberg, 2001) is adopted:

$$\begin{cases} p_K(\rho_i) = \frac{\rho_i}{1 + P(1 - \rho_i)}, \\ p_M(\rho_i) = \rho_i \end{cases} \quad (15)$$

where the penalization factor is set to $P = 4$, following previous work (Stolpe and Svanberg, 2001). From Eq. (14), one can then derive

$$\frac{\partial k(\mathbf{q})}{\partial \rho_i} = \frac{1}{2k(\mathbf{q})} \boldsymbol{\psi}^T \left[\frac{\partial \mathbf{K}(\rho_i)}{\partial \rho_i} - k^2(\mathbf{q}) \frac{\partial \mathbf{M}(\rho_i)}{\partial \rho_i} \right] \boldsymbol{\psi}. \quad (16)$$

To prevent the occurrence of checkerboard patterns during the optimization process, a filter is applied to smooth the sensitivities (Huang and Xie, 2009), as follows:

$$\tilde{\alpha}_i = \frac{\sum_{j=1}^N w_j(r_{ij}) \alpha_j}{\sum_{j=1}^N w_j(r_{ij})}. \quad (17)$$

Here, r_{ij} denotes the distance between elements i and j , and w_i is the weighting coefficient of element i , which is determined by

$$w_i = \begin{cases} r_{\min} - r_{ij} & \text{if } r_{ij} < r_{\min} \\ 0 & \text{if } r_{ij} \geq r_{\min} \end{cases}, \quad (18)$$

where $r_{\min}$ denotes the search radius of the filter, which is kept constant at $5\sqrt{2}l_e$ throughout the optimization process (Meng et al., 2015). Here, l_e denotes the element side length. Equation (18) indicates that the weighting coefficient of element i increases as its distance from element j

decreases. Moreover, the sensitivity is taken as the average of the current and previous iterations to enhance the stability of the optimization process, namely, $\tilde{\alpha}_i = 0.5(\tilde{\alpha}_i^m + \tilde{\alpha}_i^{m-1})$, where m denotes the m th iteration. Note that, because of the spatial symmetry of the unit cell, multiple degenerate eigenvalues may occur during the optimization process, which can affect the accuracy of the sensitivities. This issue can be addressed using the simultaneous treatment of degenerate eigenvalues proposed by Jensen and Pedersen (2006).

The obtained sensitivity $\tilde{\alpha}_i$ guides the volume fraction toward V_m at the m th iteration. As indicated in Eq. (9), the volume fraction is determined by the mesh density ρ_i . In this study, following the approach of Meng et al. (2015), the element density ρ_i is updated according to the following scheme:

$$\rho_i^{m+1} = \begin{cases} \min(\rho_i^m + \Delta\rho, 1) & \text{if } \tilde{\alpha}_i^m < \tilde{\alpha}_{th}^m \\ \max(\rho_i^m - \Delta\rho, \rho_{\min}) & \text{if } \tilde{\alpha}_i^m \geq \tilde{\alpha}_{th}^m \end{cases} \quad (19)$$

where $\Delta\rho$ is set to 0.1, following Meng et al. (2015), and $\tilde{\alpha}_{th}^m$ denotes the sensitivity threshold at the m th iteration. The threshold $\tilde{\alpha}_{th}^m$ is determined via the bisection method under the condition that the numerically calculated volume fraction equals the target temporary volume fraction V_m , which follows a prescribed evolutionary scheme given by $V_m = V_{m-1}(1 - ER)$. Here, V_{m-1} corresponds to the volume fraction at the previous, that is, the $(m-1)^{th}$, iteration, and ER denotes the evolution rate, which is set to 0.02 in this study. This evolutionary scheme implies that the volume fraction of the fluid phase decreases monotonically from its initial value until the iteration process terminates. Once the volume-fraction constraint is satisfied, that is, $V_m = V^*$, the convergence of the optimization process is assessed using the following criterion:

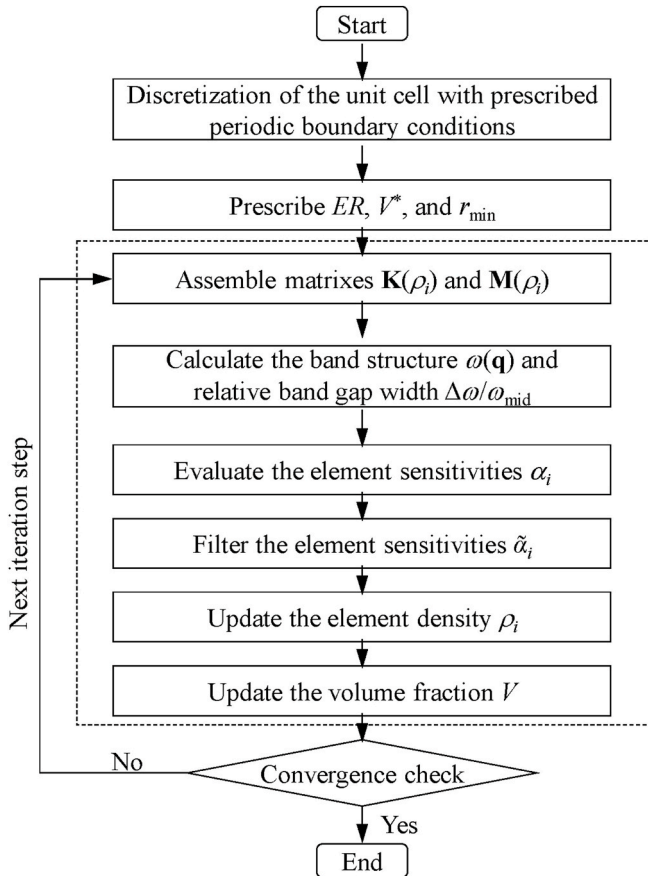


Fig. 5. Flowchart of topology optimization based on the BESO method for enhanced bandgap maximization of cylindrical array breakwaters.

$$\frac{\left| \sum_{i=1}^s (f_{m-i+1} - f_{m-s-i+1}) \right|}{\sum_{i=1}^s f_{m-i+1}} \leq \varepsilon, \quad (20)$$

where f_m denotes the value of the objective function, that is, the relative bandgap width, at the m th iteration, and ε is the convergence tolerance, taken as 10^{-4} in this study. The control parameter s is used to determine whether the relative variation in the objective function f over s consecutive iterations is smaller than ε ; here, $s = 5$. A flowchart illustrating the bandgap-optimization procedure based on the BESO method is presented in Fig. 5. Notably, several previously unspecified key parameters in the BESO method, including the target volume fraction V^* , the initial spatial distribution of the density ρ_i , that is, the initial topological configuration, and the computational grid resolution of the unit cell, can significantly influence the optimization outcome. A detailed sensitivity analysis of these parameters is presented in the following section.

4. Parameter sensitivity analysis

To ensure the robustness, reliability, and computational efficiency of the proposed BESO-based bandgap-optimization framework, a comprehensive parameter-sensitivity analysis is conducted. The analysis focuses on three key factors that may significantly influence the optimization outcome: (i) computational grid resolution, (ii) the initial topological configuration, and (iii) the prescribed target volume fraction. Before presenting these parametric studies, a representative optimization process is first examined to illustrate the convergence behavior of the algorithm.

4.1. Representative optimization process

A representative iterative topology-optimization process aimed at maximizing the width of the 1–2 ($n = 1$) bandgap is shown in Fig. 6. In this representative case, the basic geometric parameters are characterized by a still-water depth of $h = 0.50$ m and a unit cell with length and width set to $L = W = 1.0$ m. The key BESO parameters are specified as follows: a target volume fraction of $V^* = 0.60$, an initial element density of $\rho_i = 1$ for all elements, that is, the unit cell is initially fully fluid, and a computational grid of 40×40 square elements along the x- and y-axes of the unit cell. As shown in Fig. 6, the relative bandgap width $\Delta\omega/\omega_{mid}$ increases monotonically with iteration number m during the early stages

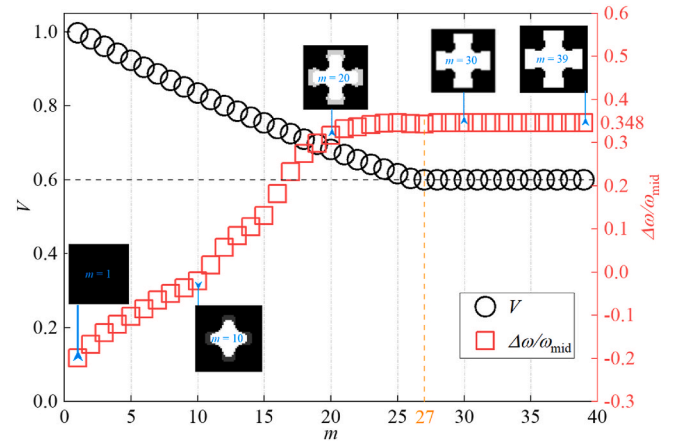


Fig. 6. Variations in the fluid-phase volume fraction V and the relative bandgap width $\Delta\omega/\omega_{mid}$ versus iteration number m . The inset figures show the unit-cell topology at five representative iteration steps: $m = 1, 10, 20, 30$, and 39 . In the inset figures, black areas indicate the fluid phase, and white areas indicate the cylinder material.

of optimization and begins to stabilize after approximately 20 iterations. Convergence is achieved at the 39th iteration, where the relative bandgap width reaches a steady value of 0.348. Meanwhile, the fluid-phase volume fraction decreases from its initial value of 1.0 and gradually approaches the prescribed target value, $V^* = 0.60$, stabilizing after approximately 27 iterations.

The evolution of the unit-cell topology further illustrates the underlying optimization mechanism. At the first iteration ($m = 1$), the unit cell consists entirely of fluid. Between the 2nd and 7th iterations, intermediate-density elements emerge and progressively evolve into solid elements. As the iterations proceed, these elements coalesce to form a distinct cross-shaped cylinder configuration. Between the 10th and 20th iterations, this configuration expands and refines, and by the 30th iteration, the topology becomes nearly invariant, indicating that the optimal bandgap configuration has been reached. This representative case demonstrates both the convergence stability and the effectiveness of the BESO algorithm in identifying a bandgap-maximizing topology. Based on this representative optimization, further parameter-sensitivity analyses are conducted to assess the influence of grid resolution, initial topology, and target volume fraction on the final optimization results, thereby providing guidance for selecting appropriate design parameters and ensuring the robustness and efficiency of the topology-optimization process.

4.2. Computational grid resolution

A grid-resolution convergence test is performed using five different uniform meshes: 20×20 , 40×40 , 50×50 , 60×60 , and 64×64 . All other parameters are kept identical to those in the representative case, namely, $h = 0.50$ m, $L = W = 1.0$ m, $V^* = 0.60$, and an initially fluid-filled unit cell. Fig. 7(a) compares the evolution of the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ with iteration number m for the five grid resolutions. The results indicate that coarse discretization (20×20) leads to noticeable deviations in the optimized bandgap width. In contrast, once the grid resolution reaches 40×40 , the convergence behavior and

final optimized bandgap width become nearly identical across all finer meshes. Although further refinement beyond 40×40 marginally improves geometric smoothness, the gain in accuracy is marginal compared with the substantial increase in computational cost. In addition, Fig. 7(b)–7(f) show the optimized unit-cell topologies corresponding to the five grid resolutions. Resolutions of 40×40 and above consistently yield the same cross-shaped topology, with only minor differences in boundary smoothness. These results indicate that the optimization outcome is mesh-independent once a sufficient resolution is achieved. By balancing accuracy and efficiency, a grid resolution of 40×40 (1600 elements) is therefore adopted for all subsequent optimization studies.

4.3. Initial topological structure

The influence of the initial spatial distribution of the density field ρ_i , that is, the initial topology, is examined using six different initial configurations: a single circular cylinder with diameters $D = 0.2L$ and $0.5L$; a single square cylinder with side lengths of $0.2L$ and $0.5L$; two circular cylinders ($D = 0.2L$ each); and two square cylinders (side length = $0.2L$ each). The remaining parameters are fixed at $h = 0.50$ m, $L = W = 1.0$ m, $V^* = 0.60$, and a grid resolution of 40×40 . Variations in the fluid-phase volume fraction V and the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ versus iteration number m are plotted in Fig. 8, together with representative intermediate topologies. In addition, to facilitate comparison and improve clarity, the key quantitative results are summarized in Table 1. The results show that, although the optimization histories differ and the numbers of iterations required for convergence vary from 43 to 51, all cases ultimately converge to a nearly identical cross-shaped topology. The final relative bandgap widths are consistently around 0.34 for all initial configurations. When considered together with the representative case shown in Fig. 6, in which the optimization starts from a fully fluid domain, these results demonstrate that the proposed BESO-based framework is largely insensitive to the initial topological configuration. This robustness is a desirable property, as it indicates that the

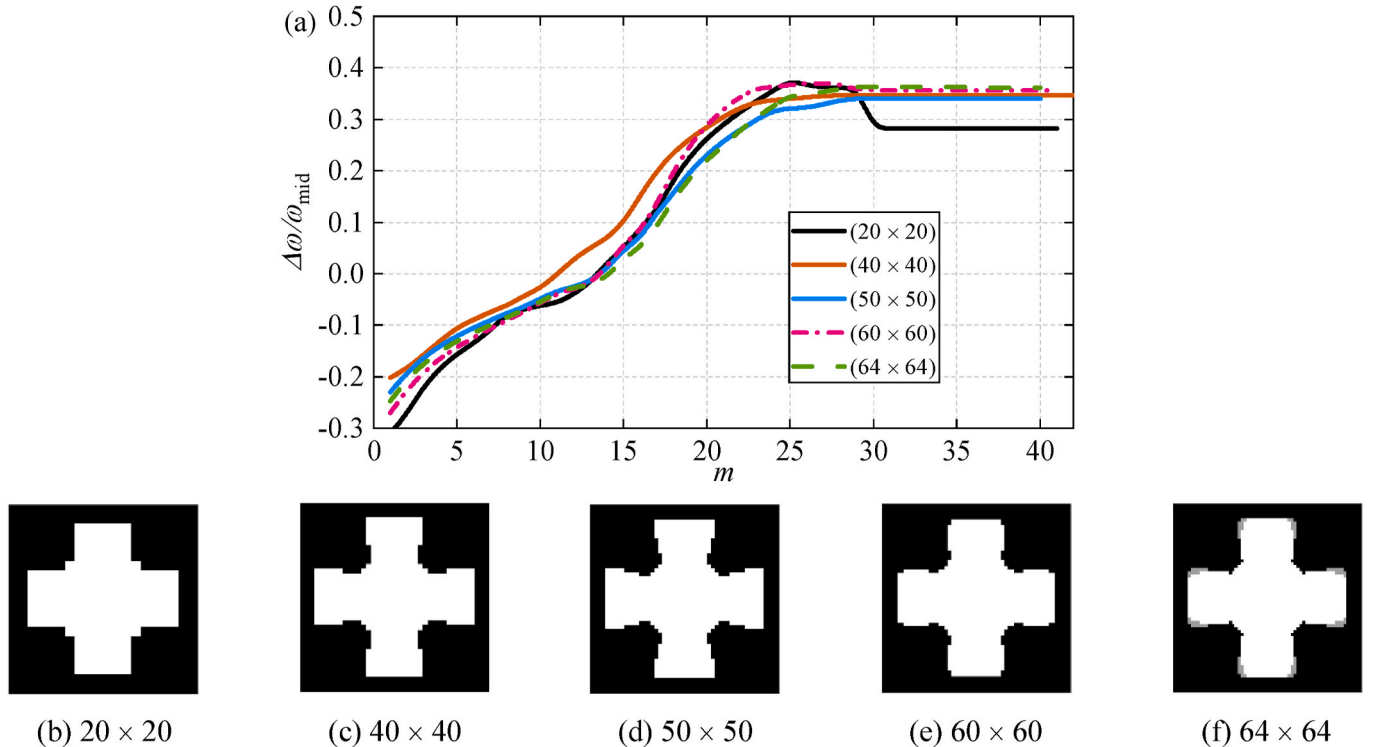


Fig. 7. (a) Variations in the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ versus iteration number m for five different grid resolutions. (b)–(f) Optimized unit-cell topologies corresponding to the five grid resolutions, where black areas indicate the fluid phase and white areas indicate the cylinder material.

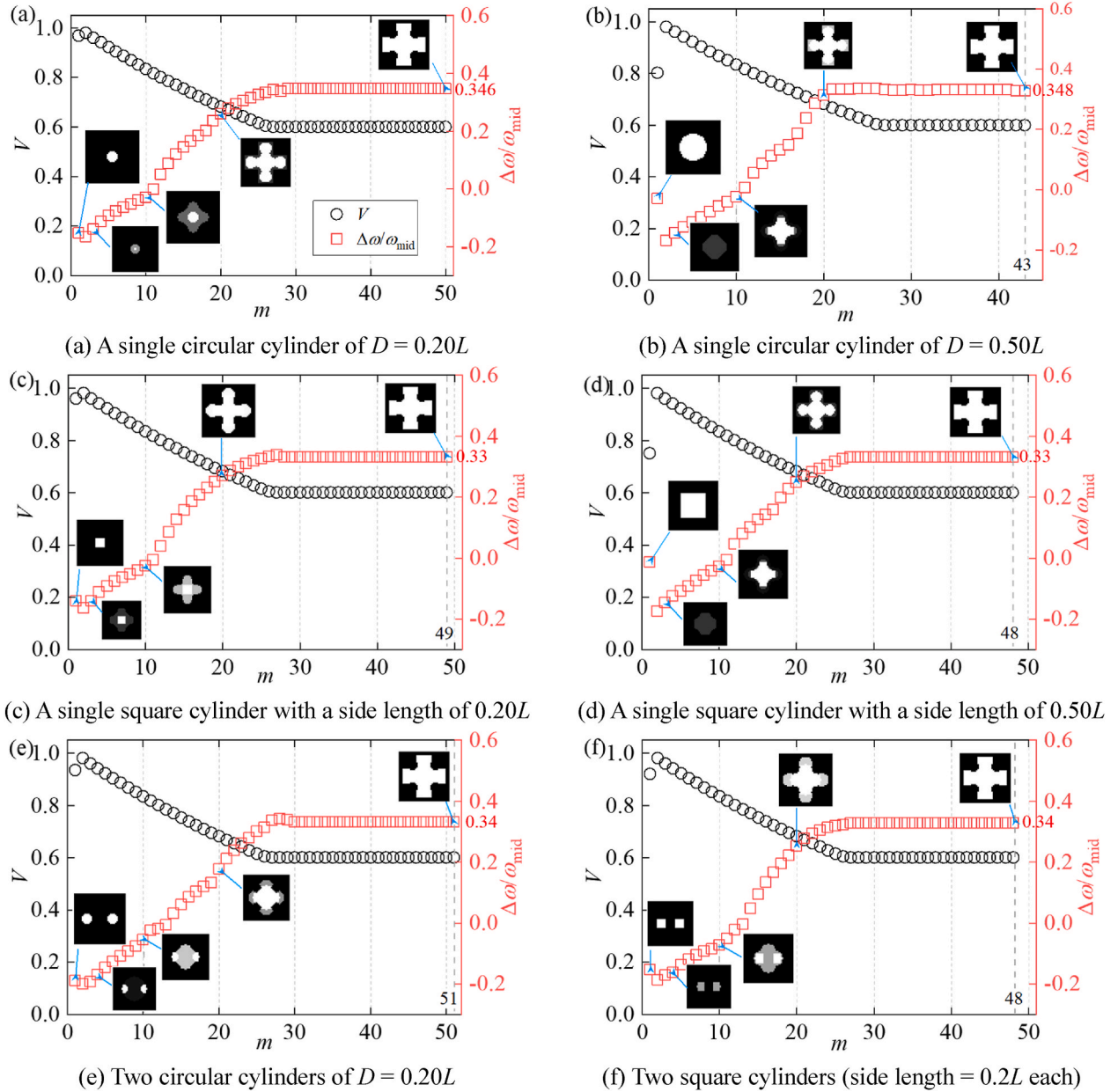


Fig. 8. Variations in the fluid-phase volume fraction V and the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ versus iteration number m for different initial topologies. The inset figures show the unit-cell topology at representative iteration steps. In the inset figures, black areas indicate the fluid phase, and white areas indicate the cylinder material.

algorithm is capable of reliably identifying the global or near-global optimum for bandgap maximization. The tabulated results clearly confirm that the optimized cross-shaped topology consistently yields the largest bandgap width across different initial configurations.

4.4. Control volume fraction

Prior to the BESO iterations, the target volume fraction V^* must be prescribed. Variations in V^* may lead to different optimization outcomes. To quantify this effect, a parametric study is conducted for V^* values ranging from 0.1 to 0.9 in increments of 0.1. Two limiting cases, $V^* = 0$ (a fully solid vertical-wall breakwater) and $V^* = 1$ (no breakwater), are excluded because they are physically trivial. All other parameters are set to $h = 0.50$ m, $L = W = 1.0$ m, an initially fluid-filled unit cell, and a 40×40 grid resolution. Variations in the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ versus iteration number m are shown in Fig. 9

for different values of V^* . The optimized bandgap widths $\Delta\omega/\omega_{\text{mid}}$ are listed in Table 2. The results reveal a strong and highly nonlinear dependence of the optimized bandgap width on the target volume fraction. For small values of V^* (for example, $V^* = 0.1$), the optimized bandgap width is negligible, indicating insufficient scattering strength. As V^* increases to 0.6, the relative bandgap width increases and reaches a maximum of 0.346. Beyond this value, further increases in V^* lead to a reduction in bandgap width, suggesting excessive blockage that suppresses favorable wave-interference mechanisms. These findings indicate that the target volume fraction is the most influential parameter in the topology-optimization process. Consequently, in practice, multiple optimization runs with different V^* values are required to identify the configuration that yields the maximum bandgap width. The sensitivity analysis confirms that the proposed optimization framework is robust with respect to numerical parameters and initial conditions, and the volume fraction plays a dominant role in determining the achievable

Table 1
Optimization results for different initial topological structures.

Initial topological structures	Iteration number m	Relative bandgap width $\Delta\omega/\omega_{\text{mid}}$	Optimization results
(a) A single circular cylinder of $D = 0.20L$	50	0.346	cross-shaped
(b) A single circular cylinder of $D = 0.50L$	43	0.348	cross-shaped
(c) A single square cylinder with a side length of $0.20L$	49	0.33	cross-shaped
(d) A single square cylinder with a side length of $0.50L$	48	0.33	cross-shaped
(e) Two circular cylinders of $D = 0.20L$	51	0.34	cross-shaped
(f) Two square cylinders (side length = $0.2L$ each)	48	0.34	cross-shaped

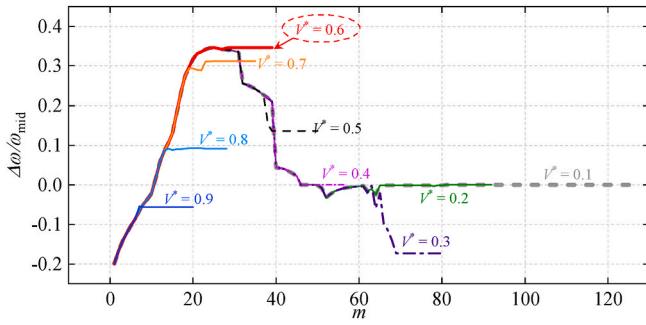


Fig. 9. Variations in the relative bandgap width $\Delta\omega/\omega_{\text{mid}}$ versus iteration number m for different control volume fractions V^* .

Table 2
Optimization results for different target volume fractions V^* .

Control volume fraction V^*	Iteration number m	Optimized relative bandgap width $\Delta\omega/\omega_{\text{mid}}$
0.1	126	-4.069E-4
0.2	92	-9.484E-4
0.3	80	-0.174
0.4	57	-0.001
0.5	50	0.136
0.6	39	0.346
0.7	35	0.312
0.8	28	0.092
0.9	20	-0.057

bandgap performance.

5. Bandgap maximization via topology optimization

Based on the parameter-sensitivity analysis presented in Section 4, this section investigates the systematic maximization of bandgaps using the proposed BESO-based topology-optimization framework. Unit-cell topologies are optimized to maximize the relative width of the n -($n+1$) bandgap for $n = 1-5$, and the resulting band structures are analyzed in detail. The optimized unit cells are then applied to infinite arrays of bottom-mounted cylinders to evaluate their wave-reflection and wave-transmission performance, thereby establishing a direct link between dispersion characteristics and hydrodynamic behavior. Unless otherwise specified, the still-water depth is fixed at $h = 0.50$ m, and the unit-cell dimensions are $L = W = 1.0$ m.

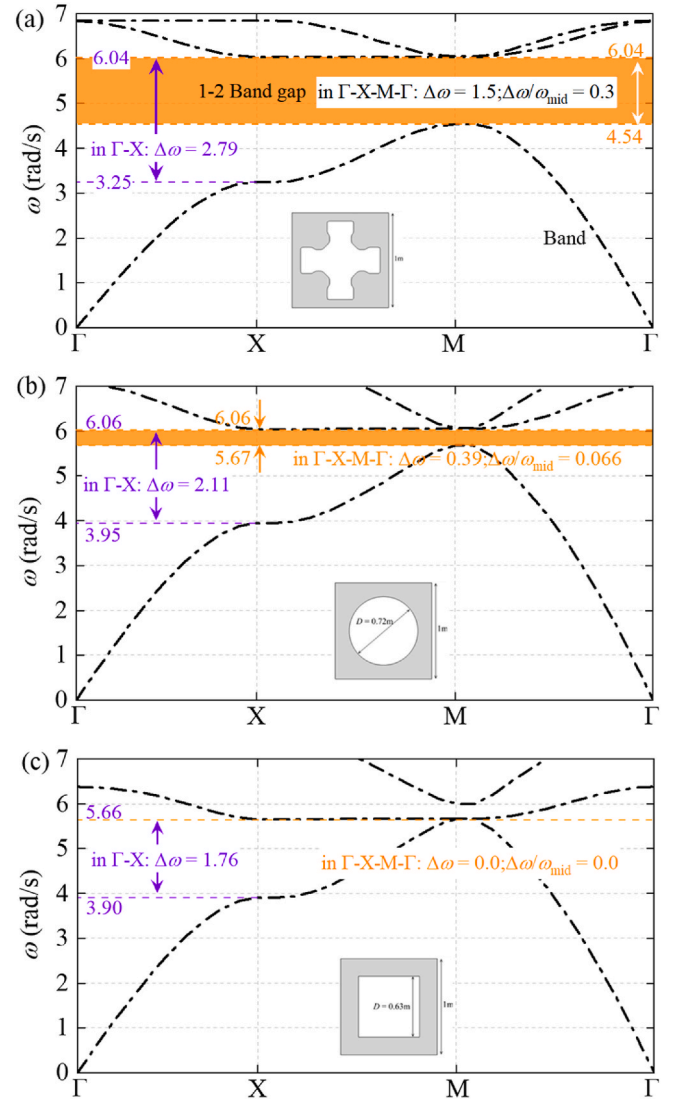


Fig. 10. Band structures along the Γ -X-M- Γ path for (a) the optimized cross-shaped topology, (b) circular cylinders, and (c) square cylinders, all with an identical control volume fraction of $V^* = 0.60$.

5.1. Bandgap enhancement for the 1-2 band

As an initial demonstration, Fig. 10(a) shows the band structure of the unit cell optimized to maximize the relative width of the 1-2 bandgap, corresponding to the topology identified in Fig. 6. A complete bandgap is observed along the entire Bloch wave-vector path Γ -X-M- Γ , with a relative bandgap width of $\Delta\omega/\omega_{\text{mid}} = 0.30$. The absolute bandgap spans from 4.54 to 6.04 rad/s, yielding a bandwidth of 1.50 rad/s. This result indicates that the optimized topology supports direction-independent wave suppression, a highly desirable property for break-water applications. In addition to the complete bandgap, a significantly

Table 3
Comparison of bandgap widths for optimized cross-shaped, circular, and square cylinders.

	Optimized cross-shaped	Circular	Square
Bandgap along Γ -X	2.79 rad/s (3.25–6.04 rad/s)	2.11 rad/s (3.95–6.06 rad/s)	1.76 rad/s (3.90–5.66 rad/s)
Bandgap along Γ -X-M- Γ	1.50 rad/s (4.54–6.04 rad/s)	0.39 rad/s (5.67–6.06 rad/s)	0 rad/s

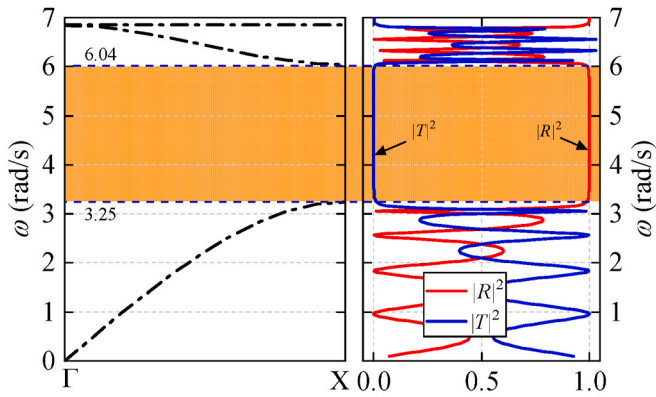
wider partial bandgap is observed along the Γ -X direction, extending from 3.25 to 6.04 rad/s. The corresponding bandgap width along Γ -X is therefore 2.79 rad/s, which is approximately 1.86 times larger than that of the complete bandgap along the Γ -X-M- Γ path. This pronounced directional enhancement suggests that the optimized cross-shaped topology interacts particularly strongly with waves propagating along the principal lattice direction, resulting in intensified scattering and mode coupling.

For comparison, Fig. 10(b) and (c) show the band structures of unit cells containing conventional circular and square cylinders, respectively, all with the same target volume fraction of $V^* = 0.60$. The corresponding bandgap characteristics are summarized in Table 3. Along the Γ -X direction, the square cylinder exhibits a bandgap width of 1.76 rad/s, with lower and upper cutoff frequencies of 3.90 and 5.66 rad/s, respectively. Similarly, the circular cylinder produces a marginally wider bandgap of 2.11 rad/s, spanning from 3.95 to 6.06 rad/s. Although both conventional configurations can generate partial bandgaps along Γ -X, their bandgap widths are considerably smaller than that of the topology-optimized cross-shaped cylinder, which achieves a bandgap width of 2.79 rad/s over the frequency range from 3.25 to 6.04 rad/s. More importantly, along the Γ -X-M- Γ path, the square and circular cylinders exhibit considerably limited bandgap effects, with bandgap widths of 0 and 0.39 rad/s, respectively. In contrast, the optimized cross-shaped topology maintains a wide bandgap of 1.50 rad/s along this path. These comparisons demonstrate that the topology-optimized cross-shaped cylinder not only significantly enlarges the bandgap width but also considerably enhances directional robustness. Such performance is particularly relevant for coastal environments, where waves typically approach from multiple directions rather than from a single dominant direction. The superior bandgap performance of the optimized topology can be attributed to its geometric complexity, which introduces multiple scattering interfaces and elongated effective wave paths within the unit cell. These features promote stronger Bragg

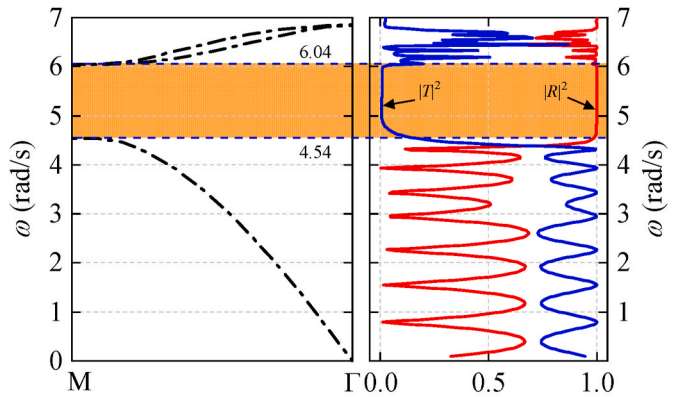
scattering and enhanced interaction among wave modes, facilitating the opening and broadening of bandgaps beyond what can be achieved with conventional simple geometries.

The bandgap characteristics revealed in the unit-cell band structures (Fig. 10) are directly reflected in the hydrodynamic response of infinite cylinder arrays. Specifically, based on the correspondence between the first Brillouin-zone boundaries and the physical-space directions, the Γ -X direction aligns with the x -axis, whereas the M- Γ direction corresponds to the oblique direction $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$. Because the unit cell employed in this study is square, the M- Γ direction represents a 45° diagonal orientation. The bandgap characteristics revealed by the unit-cell dispersion analysis are next examined at the scale of infinite cylinder arrays. Reflection and transmission coefficients are computed using the two-dimensional linear potential-flow finite-element model for waves incident along both the Γ -X and M- Γ directions. For waves propagating along the x -axis, that is, with an incident wave angle of $\theta = 0^\circ$ (Γ -X direction), the cylinder arrays are arranged at uniform spacing along the centerline of the wave flume. For obliquely incident waves along the $\theta = 45^\circ$ diagonal (M- Γ direction), the incident waves still propagate along the x -axis, and the entire cylinder array is rotated by 45° to correspond to the wave-propagation direction. The mirror-image principle (Yeung and Sphaier, 1989) is employed to approximate an infinite periodic array.

Fig. 11(a) and (b) show that, within the bandgap frequency range, the reflected wave energy, represented by the squared reflection coefficient $|R|^2$, approaches unity, whereas the transmitted wave energy, represented by $|T|^2$, is nearly zero. This result indicates that the incident wave energy is almost completely reflected by the cylinder array throughout the bandgap frequency range. This behavior is observed for both normal and oblique incidence, confirming the directionally robust wave-blocking capability of the optimized topology. The spatial distributions of normalized wave amplitude further corroborate these findings. Fig. 11(c) and (d) show the dimensionless wave-amplitude



(a) Band structure in Γ -X, squared reflection coefficient $|R|^2$, and squared transmission coefficient $|T|^2$ for wave direction $\theta = 0^\circ$



(b) Band structure along M- Γ , squared reflection coefficient $|R|^2$, and squared transmission $|T|^2$ for wave direction $\theta = 45^\circ$



(c) Normalized wave-elevation amplitude A/A_i along the wave tank at $\omega = 5.00$ rad/s and $\theta = 0^\circ$



(d) Normalized wave-elevation amplitude A/A_i along the wave tank at $\omega = 5.00$ rad/s and $\theta = 45^\circ$

Fig. 11. Band-structure diagrams, squared reflection coefficient $|R|^2$, squared transmission coefficient $|T|^2$, and dimensionless wave amplitude A/A_i for two incident wave directions under the optimized cross-shaped topology.

distribution A/A_i at $\omega = 5.00$ rad/s, a frequency located within the bandgap. In both cases, the incident waves are fully reflected upon encountering the array and do not propagate through it, demonstrating the robustness of the wave-suppression mechanism under both normal and oblique incidence. These results reveal a direct correspondence between the bandgap characteristics of the optimized unit cell and the wave-blocking performance of the cylinder array.

5.2. Optimization of higher-order bandgaps ($n > 1$)

Analogously, the proposed topology-optimization framework can be extended to target other adjacent bandgaps, that is, the n –($n+1$) bandgaps for $n = 2, 3, 4$, and 5 , corresponding in this work to the 2–3, 3–4, 4–5, and 5–6 higher-order bandgaps. The objective function and constraints in Eq. (10) are adjusted accordingly by setting n to the desired band index. The fluid-phase volume fraction V^* within the unit cell must be tuned in accordance with the procedure outlined in Section 4 to ensure effective optimization. This extension allows the BESO-based approach to systematically maximize multiple bandgaps, thereby providing greater flexibility in controlling wave propagation across a broader frequency spectrum.

The results of the higher-order bandgap maximization are shown in Fig. 12, which illustrates both the band structures and the corresponding optimized unit-cell topologies for the 2–3, 3–4, 4–5, and 5–6 bandgaps. Significant variations in the optimized cylinder geometries are observed across the different bandgaps. Following the 2–3 bandgap optimization (Fig. 12(a)), the unit-cell cylinders adopt a distinct X-shaped configuration with a target volume fraction of $V^* = 0.60$, achieving the largest relative bandgap width of $\Delta\omega/\omega_{\text{mid}} = 0.30$, bounded by lower and upper frequencies of 4.20 and 6.53 rad/s, respectively. This configuration exhibits strong symmetry and a clearly open central region, which

enhances wave scattering across the entire bandgap. For the 3–4 bandgap (Fig. 12(b)), the cylinders retain an overall X-shaped form, but with pronounced irregular extensions at the corners, and a higher target volume fraction of $V^* = 0.80$. This results in a much narrower relative bandgap width of $\Delta\omega/\omega_{\text{mid}} = 0.15$, with the frequency range spanning from 5.45 to 6.36 rad/s. The presence of corner features likely contributes to subtle local resonances but does not significantly widen the global bandgap. Following the 4–5 and 5–6 bandgap optimizations (Fig. 12(c) and (d)), the cylinders generally form a plus-shaped ("+") configuration, while irregular protrusions remain at the corners. The target volume fractions are $V^* = 0.79$ and $V^* = 0.70$ for the 4–5 and 5–6 bandgaps, respectively, resulting in relative bandgap widths of 0.24 and 0.23. The corresponding absolute frequency ranges are 5.93–7.55 rad/s and 6.28–7.89 rad/s. Compared with the 2–3 bandgap, these higher-order optimizations produce narrower relative bandgaps, but the plus-shaped topology still provides effective wave-scattering mechanisms across the designated frequency ranges.

Overall, the 2–3 bandgap maximization yields the largest relative bandgap, whereas the 3–4 bandgap produces the narrowest, highlighting the sensitivity of the optimized unit-cell topology to the targeted bandgap. The geometric features, including X-shaped and plus-shaped forms as well as corner irregularities, are clearly tailored to enhance wave attenuation within specific frequency ranges, demonstrating the capability of the BESO-based topology optimization to adaptively engineer unit-cell configurations for multiple bandgap targets.

Importantly, the unit cells optimized for different bandgaps exhibit complementary bandgap distributions in the frequency domain. Specifically, as shown in Fig. 12(a), the unit cell optimized for the 2–3 bandgap generates two separated bandgaps located at 4.20–6.53 rad/s and 7.77–8.21 rad/s. In contrast, the unit cell optimized for the 5–6 bandgap (Fig. 12(d)) produces three distinct bandgaps spanning

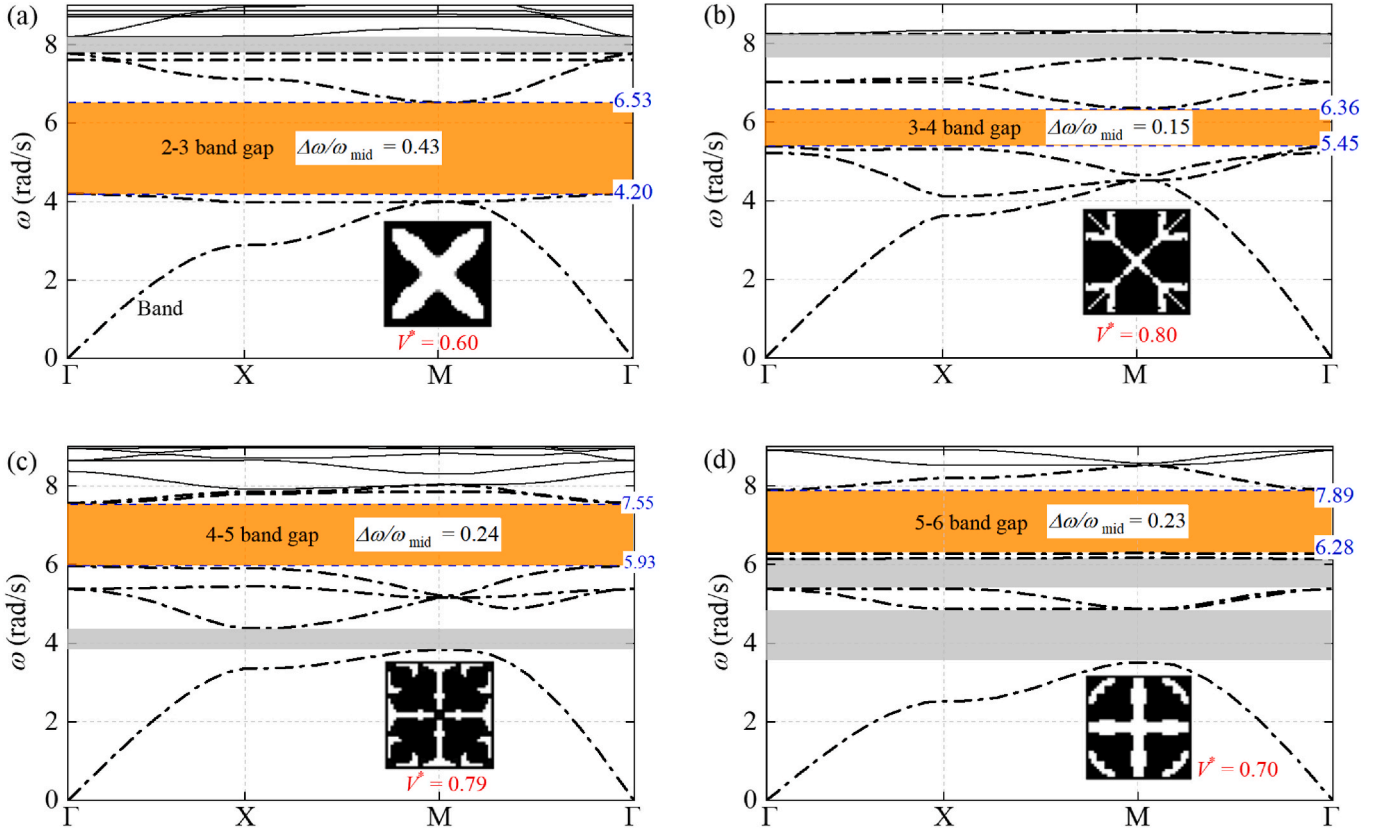


Fig. 12. Band structures of the optimized unit-cell topologies for (a) the 2–3 bandgap with $V^* = 0.60$, (b) the 3–4 bandgap with $V^* = 0.80$, (c) the 4–5 bandgap with $V^* = 0.79$, and (d) the 5–6 bandgap with $V^* = 0.70$. The inset figures illustrate the corresponding optimized unit-cell topologies, where the black regions denote the fluid phase and the white regions denote the cylinder material.

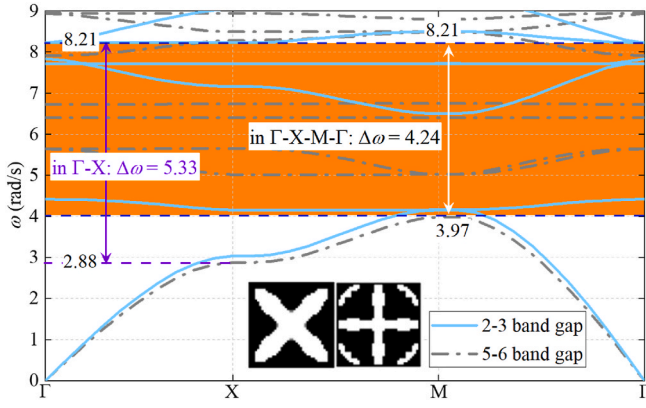
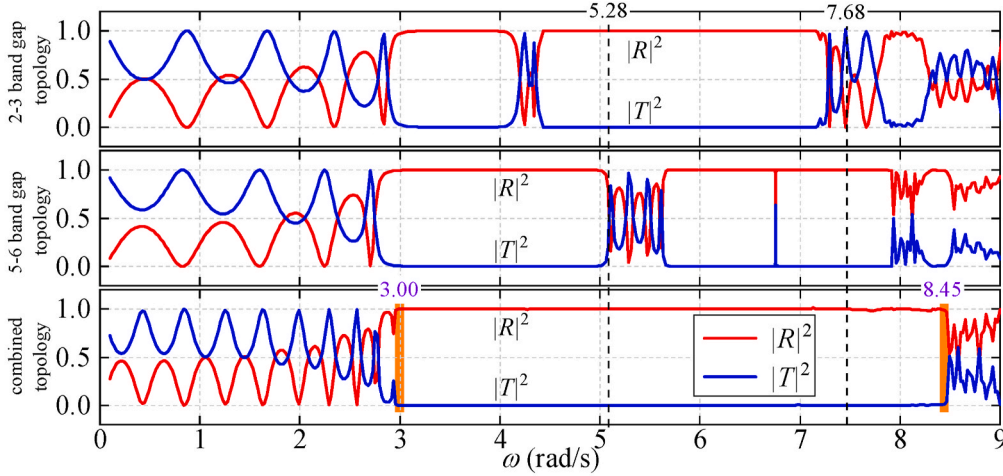


Fig. 13. Band structures of the combined configuration consisting of the unit-cell topology optimized for the 2–3 bandgap and that optimized for the 5–6 bandgap.

3.52–4.85 rad/s, 5.38–6.13 rad/s, and 6.28–7.89 rad/s. These observations suggest that different optimized topologies can open bandgaps over partially overlapping yet complementary frequency ranges, thereby providing an opportunity for further bandgap broadening through a combination strategy.

Motivated by this observation, a combined configuration is proposed by arranging the unit cell optimized for the 2–3 bandgap together with that optimized for the 5–6 bandgap. The schematic configuration and the corresponding band structure are shown in Fig. 13. After combining the two optimized unit-cell topologies, the individual bandgaps merge and overlap, forming a substantially broadened effective bandgap. Along the complete Bloch wave-vector path Γ -X-M- Γ , a wide bandgap is obtained between 3.97 and 8.21 rad/s, yielding an absolute bandgap width of $\Delta\omega = 4.24$ rad/s, as highlighted by the orange-shaded region in Fig. 13. An even more pronounced bandgap widening is observed along the Γ -X direction, where the lower and upper cutoff frequencies extend to 2.88 and 8.21 rad/s, respectively. This corresponds to a bandgap width of $\Delta\omega = 5.33$ rad/s, which is significantly larger than that along the Γ -X-M- Γ path. The enhanced bandgap along Γ -X indicates that the



(a) Squared reflection coefficients $|R|^2$ and squared transmission coefficients $|T|^2$



(b) Normalized wave-elevation amplitude along the wave tank at $\omega = 5.28$ rad/s



(c) Normalized wave-elevation amplitude along the wave tank at $\omega = 7.68$ rad/s

Fig. 14. Squared reflection coefficients $|R|^2$ and squared transmission coefficients $|T|^2$, together with the dimensionless wave amplitude A/A_i , for arrays of cylinders with cross-shaped topologies optimized for the 2–3 bandgap, the 5–6 bandgap, and their combined topology.

combined topology is particularly effective in suppressing wave propagation along the principal lattice direction, while still maintaining a wide and robust bandgap for all propagation directions. Quantitative analysis demonstrates that the combined configuration of the 2–3- and 5–6-bandgap-optimized unit cells results in a substantial enhancement in bandgap width compared with the individual optimized cases. Specifically, the effective bandgap width achieved by the combined strategy is approximately 1.8 and 2.6 times larger than the maximum bandgap widths obtained from the standalone 2–3 and 5–6 bandgap optimizations, respectively. These results demonstrate that the proposed combination strategy provides a simple yet powerful approach for broadband bandgap engineering, enabling effective wave attenuation over a much wider frequency range without introducing additional geometric complexity at the unit-cell level.

Fig. 14 provides a comprehensive assessment of the wave-attenuation performance of cylinder arrays optimized for the 2–3 bandgap, the 5–6 bandgap, and their combined configuration by examining both the frequency-domain reflection and transmission characteristics and the corresponding spatial wave-field distributions. As shown in Fig. 14(a), the combined topology exhibits an exceptionally stable response over a wide frequency range. Specifically, within the interval from 3.00 to 8.45 rad/s, the squared reflection coefficient $|R|^2$ remains equal to unity, while the squared transmission coefficient $|T|^2$ remains essentially zero. This behavior indicates that the incident wave energy is completely reflected across this entire frequency band, demonstrating broadband and near-perfect wave-blocking capability. In contrast, the arrays optimized individually for the 2–3 and 5–6 bandgaps show pronounced frequency-selective behavior. At $\omega = 5.28$ rad/s, which lies within the 2–3 bandgap but outside the effective bandgap of the 5–6-optimized array, the 2–3-bandgap topology achieves almost total reflection, with $|R|^2 \approx 1$, whereas the 5–6-bandgap topology allows substantial wave transmission, with $|R|^2$ approaching zero. Conversely, at $\omega = 7.68$ rad/s, the 5–6-bandgap-optimized array exhibits strong reflection, whereas the 2–3-bandgap-optimized array becomes ineffective. These observations indicate that each individually optimized topology provides effective wave attenuation only within its targeted bandgap and fails outside that frequency range. More importantly, at both $\omega = 5.28$ rad/s and 7.68 rad/s, frequencies at which neither the 2–3- nor the 5–6-bandgap-optimized array alone can block wave propagation over the entire array length, the combined configuration consistently maintains $|R|^2 \approx 1$ and $|T|^2 = 0$. This result highlights the complementary and synergistic interaction between the two optimized unit-cell topologies. Frequency components that fall within the pass-band of one topology are effectively suppressed by the bandgap of the other, leading to a continuous and significantly broadened effective stopband.

The spatial distributions of the normalized wave amplitude further substantiate these findings. Fig. 14(b) and (c) show the dimensionless wave-elevation amplitude A/A_i along the wave flume at $\omega = 5.28$ rad/s and 7.68 rad/s, respectively. For the single-bandgap arrays, partial wave penetration or standing-wave patterns are observed downstream of the array when the incident frequency lies outside the corresponding bandgap. In sharp contrast, the combined topology consistently prevents wave propagation beyond the array, with the wave field confined to the upstream region and exhibiting clear reflection-dominated patterns. These spatial results provide direct physical evidence that the broadened bandgap achieved through topology combination translates into robust suppression of wave energy in the physical domain. Thus, Fig. 14 establishes a clear and quantitative link between bandgap widening at the dispersion level and enhanced hydrodynamic performance at the array scale. By combining unit cells optimized for different bandgaps, the proposed approach effectively overcomes the inherent narrowband limitation of single-topology designs and achieves broadband, directionally robust wave attenuation. This capability is particularly advantageous for coastal-protection applications, in which incoming waves span wide frequency ranges and vary dynamically over time,

underscoring the practical engineering potential of the proposed bandgap-combination strategy.

Taken together, this section has demonstrated a systematic pathway from bandgap-oriented topology optimization at the unit-cell level to enhanced wave-attenuation performance at the scale of practical cylindrical breakwater arrays. Starting from the maximization of individual bandgaps using the proposed BESO-based framework, a series of optimized unit-cell topologies were obtained for different adjacent bands, revealing that distinct geometric features are required to effectively open and control bandgaps over different frequency ranges. Building on these results, a combined configuration composed of unit cells optimized for the 2–3 and 5–6 bandgaps was further proposed to overcome the inherently narrowband nature of single-topology designs. The band-structure analysis confirms that the combination of two complementary optimized topologies leads to pronounced overlap and merging of multiple bandgaps, resulting in a substantially widened effective stopband. More importantly, hydrodynamic simulations of the corresponding cylinder arrays establish a direct and quantitative connection between the broadened bandgap and the actual wave-blocking performance. The combined array achieves near-complete reflection and negligible transmission over a broad frequency interval from 3.00 to 8.45 rad/s, significantly exceeding the effective attenuation ranges of arrays based on individually bandgap-optimized topologies. The spatial distributions of wave elevation further corroborate this finding, showing that incident wave energy is consistently confined to the upstream region of the array across the entire target frequency range. These results highlight the effectiveness and robustness of the proposed bandgap-combination strategy in translating dispersion-level optimization into tangible hydrodynamic benefits. From an engineering perspective, the ability to achieve broadband and directionally robust wave attenuation using periodically arranged cylinders offers a promising design paradigm for compact and efficient coastal-protection structures. The findings of this section provide a crucial link between topology-optimization theory and practical breakwater design, paving the way for the development of next-generation metamaterial-inspired coastal defenses capable of mitigating waves over a wide spectrum of realistic sea states.

6. Conclusions

This study has established an integrated computational framework that synergizes band-structure analysis, topology optimization, and hydrodynamic validation for the design of periodic structures with tailored wave-blocking capabilities. A two-dimensional Bloch-wave band-structure model is first developed to characterize wave propagation in infinite periodic cylinder arrays. Comparisons with frequency-domain potential-flow simulations confirm that the predicted bandgaps accurately correspond to full-reflection frequency intervals while offering significantly improved computational efficiency. Building on this model, a BESO framework is developed to maximize the relative width of prescribed bandgaps by configuring the unit-cell topology. Parameter-sensitivity analyses demonstrate that the optimization results are robust with respect to grid resolution and initial topology, whereas the target fluid-phase volume fraction plays a dominant and highly nonlinear role in determining the achievable bandgap width. Compared with conventional circular and square cylinders, the topology-optimized configurations exhibit substantially wider bandgaps and improved directional robustness. The framework is further extended to higher-order bandgaps, revealing that distinct geometric features are required to effectively open bandgaps over different frequency ranges. Importantly, by combining unit cells optimized for complementary bandgaps, multiple separated bandgaps can be merged into a significantly broadened effective stopband. Hydrodynamic simulations of cylinder arrays demonstrate that this combined configuration achieves near-complete wave reflection over a wide frequency interval, far exceeding the attenuation capability of arrays based on single optimized topologies.

The results establish a clear and quantitative link between bandgap optimization at the unit-cell level and broadband wave attenuation at the array scale. The proposed topology-optimization and bandgap-combination strategy provides an effective and practical design paradigm for compact, broadband, and directionally robust breakwater structures, offering new opportunities for metamaterial-inspired coastal-protection applications.

Although the present framework is developed within an idealized two-dimensional and linear wave context, it provides a physically grounded basis for practical engineering design. The optimized unit-cell topologies can be interpreted as conceptual prototypes for compact and broadband-efficient breakwaters. In practical applications, the proposed bandgap-oriented topology-optimization framework can be further extended to incorporate three-dimensional effects, irregular wave conditions, viscous effects (Zhou and Song, 2026), and constructability constraints, thereby enabling its integration into real-world coastal-protection design. In addition, the bandgap-combination strategy introduced here may be further developed toward graded or adaptive breakwater designs, enabling tailored wave attenuation over site-specific and time-varying wave spectra. Experimental validation through physical model tests is also desirable to assess scale effects and practical performance. In a broader context, this work provides a generalizable and efficient paradigm for the topology optimization of phononic-crystal-like structures in fluid media. The developed framework is not limited to water-wave breakwaters but can also be adapted to other wave-control problems involving acoustics, elasticity, or optics

in periodic settings, wherever bandgap engineering is desired.

CRedit authorship contribution statement

Zhongbing Zhou: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition. **Zhiwei Song:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Two-dimensional linear potential-flow finite-element model in the frequency domain

To determine the complex velocity potential $\phi(x, y)$, which satisfies the Helmholtz equation (Eq. (1)), the total potential $\phi(x, y)$ is decomposed into an incident component $\phi_I(x, y)$ and a diffraction component $\phi_D(x, y)$:

$$\phi(x, y) = \phi_I(x, y) + \phi_D(x, y). \quad (\text{A.1})$$

The linear incident velocity potential is prescribed as $\phi_I(x, y) = Ae^{ik_x \cos \theta + ik_y \sin \theta}$, where A is the incident wave amplitude, and θ denotes the wave-propagation direction measured from the x -axis. Substituting Eq. (A.1) into Eq. (1) yields the governing Helmholtz equation for the diffraction velocity potential $\phi_D(x, y)$:

$$\nabla^2 \phi_D(x, y) + k^2 \phi_D(x, y) = 0. \quad (\text{A.2})$$

By invoking the mirror-image principle (Yeung and Sphaier, 1989), the diffraction problem for a row of vertically mounted cylinders located at the center of a wave tank with vertical sidewalls can be equivalently transformed into that of an infinite array of identical cylinders in an unbounded domain. On the surfaces of both the vertical sidewalls and the mounted cylinders, the impermeable boundary condition is imposed on the diffraction potential ϕ_D :

$$\frac{\partial \phi_D}{\partial \mathbf{n}} = -\frac{\partial \phi_I}{\partial \mathbf{n}}, \quad (\text{A.3})$$

where $\mathbf{n}$ denotes the outward unit normal vector on the boundary. At infinity, the Sommerfeld radiation condition is enforced to ensure outgoing waves:

$$\lim_{r \rightarrow \infty} r^{1/2} \left(\frac{\partial \phi_D}{\partial r} - ik \phi_D \right) = 0, \quad (\text{A.4})$$

with $r = (x^2 + y^2)^{0.5}$. Solving Eq. (A.2) subject to the boundary conditions in Eq. (A.3) and (A.4) yields the diffraction potential ϕ_D . The total velocity potential $\phi(x, y)$ is then obtained from Eq. (A.1), and the free-surface elevation $\eta(x, y)$ is evaluated as $\eta = i\omega \phi(x, y)/g$. The reflection coefficient R and transmission coefficient T are subsequently computed based on the wave-energy flux as

$$R = \sqrt{P_D/P_I}, T = \sqrt{P/P_I}, \quad (\text{A.5})$$

with $P_D = \int_S -\frac{\rho\omega}{2} \text{Im} \left(\phi_D \frac{\partial \phi_D^*}{\partial x} \right) dS$, $P_I = \int_S -\frac{\rho\omega}{2} \text{Im} \left(\phi_I \frac{\partial \phi_I^*}{\partial x} \right) dS$, $P = \int_S -\frac{\rho\omega}{2} \text{Im} \left(\phi \frac{\partial \phi^*}{\partial x} \right) dS$, where ρ denotes the fluid density, and ϕ^* , ϕ_D^* , and ϕ_I^* are the complex conjugates of the complex velocity potentials ϕ , ϕ_D , and ϕ_I , respectively.

In the present study, the above mixed boundary-value problem is solved using the finite-element method. Compared with the boundary-element method commonly adopted in wave-structure interaction problems, the finite-element method is better suited for handling complex geometries and large-scale computational domains. After discretizing the fluid domain into a finite number of elements, the diffraction potential within each element

is approximated using bilinear shape functions:

$$\phi_D = \mathbf{N}\Phi_D, \quad (\text{A.6})$$

where $\mathbf{N}$ denotes the shape-function matrix, and Φ_D is the nodal diffraction-potential vector. Applying the standard Galerkin weighted-residual method to Eq. (A.2), together with the boundary conditions in Eq. (A.3) and (A.4), leads to the following weak form:

$$\int_{\Omega} \nabla \mathbf{N}^T \nabla (\mathbf{N}\Phi_D) d\Omega - k^2 \int_{\Omega} \mathbf{N}^T (\mathbf{N}\Phi_D) d\Omega = \oint_{\Gamma} \mathbf{N}^T \nabla (\mathbf{N}\Phi_D) \cdot \mathbf{n} ds. \quad (\text{A.7})$$

After incorporating the boundary conditions, the resulting linear system can be written as

$$\mathbf{A}\Phi_D = \mathbf{B}, \quad (\text{A.8})$$

with

$$\mathbf{A} = (\mathbf{D} - k^2 \mathbf{M} - \mathbf{T}_2), \mathbf{B} = \mathbf{T}_1, \mathbf{D} = \int_{\Omega} \nabla \mathbf{N}^T \nabla \mathbf{N} d\Omega, \mathbf{M} = \int_{\Omega} \mathbf{N}^T \mathbf{N} d\Omega, \mathbf{T}_1 = \oint_{\Gamma} \mathbf{N}^T \left(-\frac{\partial \phi_1}{\partial \mathbf{n}} \right) \cdot \mathbf{n} ds, \mathbf{T}_2 = \oint_{\Gamma} \mathbf{N}^T (-ik\mathbf{N}\Phi_D) \cdot \mathbf{n} ds.$$

The above linear system is solved using a preconditioned conjugate gradient method to obtain the diffraction potential ϕ_D . The present frequency-domain model, based on the Helmholtz equation, is applicable to wave interactions with bottom-mounted structures of uniform cross-section. Further validation of the model for wave scattering by cylindrical arrays can be found in the authors' previous work (Zhou and Song, 2025).

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